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Term Project: Instant Messaging App

Feature Document

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1 Introduction

This document will outline the features, scope, and vision of the instant messaging application. The product is a real time chat system consisting of a server and multiple clients that allows its users to communicate through chat rooms. This document outlines the goals, intended users, major features, operating environment, and constraints. It will serve as a foundation for later design, architecture, and testing documentation which will help guide the development process throughout its lifecycle.

1.1 Purpose

To clearly define the scope and vision of the instant messaging application. By outlining goals, features, and constraints early, this document will help ensure consistent understanding among developers and stakeholders.

1.2 Target Audience

- The development team responsible for implementing the application
- Dr. Fei Xie and TA Huan Wu
- Future developers who may extend or maintain the application

1.3 Terms and Definitions

Client: A user-facing app that connects to the chat server in order to send and receive messages.
Server: The backend system responsible for handling client connections, authentication, and messages.
Real-Time Messaging: The ability to send and receive messages without requiring refreshes.
Authentication: Verifying a user's identity through login and registration.
API: A set of endpoints that allow the frontend and backend to communicate.

2 Product Overview

WHAT: A web based instant messaging app that supports user authentication, chat room management, and real time messaging.
WHO: Students and general users who want a simple, reliable platform for group-based communication
WHY: To provide a real-time communication tool

2.1 Product Vision

For users who want a simple and reliable way to communicate in real time,
Who need organized chat rooms and conversation history,
The Instant Messaging App
Is a real-time communication platform
That enables secure user authentication, chat room management, and instant messaging,
Unlike traditional messaging tools that are feature-heavy,
Our product focuses on essential messaging features with a clean UI.

2.2 Product Scope

<u>User Management</u>

- User registration, login and logout
- Set/update first and last name
- Authentication error handling

<u>Chatroom Management</u>

- Create/delete chatrooms
- View/navigate available chatrooms

<u>Messaging</u>

- Send/receive messages in real time
- Display conversation history
- Support multiple users in a chatroom

Frontend Features

- Responsive UI
- Error messages for invalid actions/connection issues

Backend Features

- Client connection handling
- Message routing/delivery
- Data persistence for users, messages, and chatrooms

2.3 Operating Environment

The Instant Messaging App will operate in a web-based environment which requires a modern web browser, a server environment that is capable of running Node.js, internet connection for real-time communication and a database system for data storage.

2.4 Design and Implementation Constraints

Frontend

- React (JavaScript) for building the UI

Backend

- Node.js with Express for handling server logic and APIs

Database

- MongoDB for storing user credentials, messages, and chatroom data

Version Control

- Git and GitHub for agile development

Constraints

- Project timeline can limit complexity/number of features that can be implemented
- Real time communication must be implemented within the limitations of web technologies

3 User Stories

Understanding users' goals and pain points is critical for building a product that should be both usable and valuable. User stories will emphasize what users really want to accomplish, and why those goals matter. A good amount of effort should be spent on understanding users' needs in order to not implement unnecessary features or confusing workflows.

3.1 Personas

1) Valente Basilio Pena

Valente is a 21 year old, third-year computer science student at Portland State University. He lives off-campus with his family and spends a lot of his time balancing class, work, and leading a club. Valente has been using computers and smartphones since elementary school. He is currently enrolled in several computer science courses that require frequent group collaboration and often relies on online communication to share information. He is comfortable using modern web applications and messaging platforms and considers himself technically proficient. Valente is interested in using this instant messaging application because it allows him to organize conversations by course or study group, and he values real-time communication and the ability to view conversation history.

2) Antuniuh Dang

Antuniuh is a 17 year old early college student majoring in business administration. Antuniuh commutes to campus from a nearby town and often collaborates with classes remotely due to schedule conflicts. Antuniuh has completed several group-heavy courses that require teamwork, coordination, and regular communication. While Antuniuh is comfortable using common apps such as email or messaging apps, he does not have a strong technical background. Antuniuh's educational background focuses more on management and collaboration. Antuniuh is interested in this product because it provides a simple way to communicate without unnecessary complexity.

3) Adreante Beavers

Adreante Beavers, age 35 recently graduated with a degree in sociology and is working in a community outreach role. Adreante lives in a shared apartment with 3 kittens and uses technology mainly for communication and daily tasks rather than professional/technical work. Adreante has a college education but limited technical experience beyond common messaging platforms, and often feels frustrated when apps require complicated setup or unclear error messages. Adreante is interested in using this instant messaging app because it provides a straightforward way to join group conversations without advanced technical knowledge.

3.2 User Scenarios

3.2.1 <User Scenarios with Persona 1>

Valente is working on a group programming assignment for one of his computer science classes. He creates a messaging group for the project, and adds his project group members to it. Over the course of the week, he messages information about the work that was completed, the times at which meetings occurred, and links to where the work can be found. He reviews past messages to recall decisions made earlier and ensures that everyone stays aligned.

3.2.2 <User Scenarios with Persona 2>

Antoniuh often collaborates online with his classmates due to travel and scheduling issues. He is a member of a group chat that his classmates created, and Antoniuh uses it to assist him in managing when work is to be completed. He reviews the chat for past messages to ensure he knows what has already been covered and responds with short messages to confirm what he needs to be working on. Antoniuh desires a simple and easy to use system, so he can focus on collaboration rather than learning complex features.

3.2.3 <User Scenarios with Persona 3>

Adreante uses the messaging app to message co-workers and volunteers for community activities. He participates in group chats to disseminate information about events and answers questions. Adreante looks back through previous conversations to find

information about upcoming activities. He appreciates clear, straightforward interactions because he becomes frustrated when applications are confusing or provide unclear feedback.

3.3 Epic User Stories

Write at least three epic (Chapter 3 p28 3rd bullet point) user stories (Chapter 3 p27-29) with acceptance criteria in the following template. Try to define each small user story specifically so that they could directly be used in the sprint backlog.

3.3.1 <Epic User Stories 1>

Epic 1: Account Access and Identity
As a chat app user ***I want to*** create and manage an account, sign in, and control my display identity ***So that I can*** securely use the chat app and be recognizable to others.
Acceptance Criteria ***Given*** I am not signed in ***When*** I submit valid account information ***Then*** an account is created and I am signed in.
<Breakdown the features to a single feature that can be implemented in a sprint> User story 1: As a new user, I want to create an account so that I can start using the messaging app.
User story 2: As a returning user, I want to sign in so that I can access my existing conversations.
User story 3: As a user, I want to update my display name and profile picture so that others can recognize me in conversations.

3.3.2 <Epic User Stories 2>

Epic 1: Messaging and conversations
As a chat app user ***I want to*** start conversations and send messages to individuals or groups ***So that I can*** communicate with classmates, coworkers, and friends
Acceptance Criteria ***Given*** I am signed in ***When*** I start a conversation with another user ***Then*** a direct conversation is created/opened.
<Breakdown the features to a single feature that can be implemented in a sprint> User story 1: As a user, I want to start a direct conversation so that I can communicate privately with another user.
User story 2: As a returning user, I want to see message history so that I can see what was talked about previously.
User story 3: As a user, I want to create a group conversation and add members so that I can communicate with multiple people at once.

3.3.3 <Epic User Stories 3>

Epic 1: Conversation organization and Attention management

As a chat app user ***I want to*** manage unread messages and control conversation alerts ***So that I can*** focus on important conversations and reduce distractions
Acceptance Criteria ***Given*** I am not viewing a conversation ***When*** a new message arrives ***Then*** the conversation is marked as unread ***Given*** I open an unread conversation ***When*** I view the messages ***Then*** the unread status is cleared
<Breakdown the features to a single feature that can be implemented in a sprint> User story 1: As a user I want to see unread message indicators so that I know which conversations need my attention.
User story 2: As a coworker, I want to mute a conversation so that I can reduce interruptions.
User story 3: As a user I want to unmute a conversation so that I can receive message alerts again.

4 Feature List

Feature 1: Sign up	**Priority: High**	**Effort Required:** 1
Description: Allow a new user to create an account with required information and begin using the app.		
Constraints: ● Must reject invalid or incomplete registration information ● Must prevent duplicate accounts based on chosen identifier		
Comments: Make it intuitive so that non-technical users don't have any struggles.		

Feature 2: Sign in	**Priority: High**	**Effort Required:** 1
Description: Allow an existing user to authenticate and access their account and conversations.		
Constraints: ● Must deny access on invalid credentials ● Must show clear error feedback		
Comments: Reliability matters more than adding extra features (social media sign in)		

Feature 3: Sign out	**Priority: High**	**Effort Required:** 0.5
Description: Allow a signed-in user to end their session		

Constraints: ● Must return the user to the login screen (or a signed out screen)
Comments: Important for shared devices to maintain privacy

Feature 4: List of conversations	**Priority: High**	**Effort Required:** 1
Description: Show the user all the conversations they are a part of so they can choose where to communicate		
Constraints: ● Must display only conversation the user belongs to		
Comments: Landing page for app.		

Feature 5: Start a new conversation	**Priority: High**	**Effort Required:** 1
Description: Allow the user to begin a private conversation with another user		
Constraints: ● Must prevent starting a conversation with an non-existent user ● Must avoid creating duplicate DMs for the same pair of users.		
Comments:		

Feature 6: Start a group conversation	**Priority: High**	**Effort Required:** 1.5
Description: Allow the user to create a group conversation for multiple participants		
Constraints: ● Group must be able to be renamed (default name is all usernames in group) ● Creator must become a participant automatically		
Comments: Users can be in multiple group conversations, including ones that contain users they already share group conversations with.		

Feature 7: Add members to a group conversation	**Priority: High**	**Effort Required:** 1.5
Description: Allow a group participant to add additional users to the group		
Constraints: ● Must prevent adding users who are already members ● Display error message if user is unable to be added (invalid user, etc.)		
Comments:		

Feature 8: Send a message	**Priority: High**	**Effort Required:** 2
Description:		

Allow the user to send a message in a conversation
Constraints: ● Must not send empty messages ● Must attach sender identity and timestand to each message
Comments: Will require lots of tests

Feature 9: View message history	**Priority: High**	**Effort Required:** 1.5
Description: Allow the user to open a conversation and read past messages in order		
Constraints: ● Must preserve message order ● Must show messages only to conversation members		
Comments:		

Feature 10: Unread indicators	**Priority: High**	**Effort Required:** 1.5
Description: Show which conversations have new/unread messages so the user can prioritize attention		
Constraints:		

● Must mark a conversation as unread when a new message arrives and the user isn't already viewing it ● Must clear unread status when the user views the conversation
Comments:

Feature 11: Mute a conversation	**Priority: Medium**	**Effort Required:** 1
Description: Allow the user to mute a conversation to reduce distractions		
Constraints: ● Conversations that are muted should still be accessible and accumulate unread messages. ● Muting should be reversible		
Comments:		

Feature 12: Edit user profile	**Priority: Medium**	**Effort Required:** 1.5
Description: Allow the user to manage display identity that is visible in chats		
Constraints: ● Must validate profile fields ● Must persist changes for future messages and displays		

Comments:

Feature 13: Leave a group conversation	**Priority:** **Medium**	**Effort Required:** 1
Description: Allow the user participating in a group conversation to leave the group chat		
Constraints: ● After leaving, the conversation should no longer appear in their conversation list ● Must prevent leaving from breaking access control (ex-members can't read/send) ● Remaining users should be unaffected ● Display message to members in the group conversation that a member has left		
Comments:		

Feature 14: Remove a member from a group conversation	**Priority:** **Low-Medium**	**Effort Required:** 1.5
Description: Allow a group member to remove a participant who should no longer be included		
Constraints: ● This feature should only be accessible to the creator of the group conversation ● Removed users must lose access to the conversation ● Action should provide clear feedback (success/fail for the user who is removing)		
Comments:		

Feature 15: Search conversations or messages	**Priority: Low**	**Effort Required:** 2
Description: Allow the user to find specific conversations or message content from earlier discussions		
Constraints: ● Search results must respect access control ● Search results must be relevant to user input		
Comments:		

Feature 16: Error feedback	**Priority: Medium**	**Effort Required:** 1
Description: Provide the user with clear, human readable feedback when actions fail		
Constraints: ● Errors must be understandable to non-technical users ● Must not expose sensitive information in error messages		
Comments:		